

Thermal-guided Semantic Segmentation

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Public-data reconstruction: 2026

Abstract

I adapted an RGB-D segmentation workflow to RGB-thermal inputs during a SERES internship. This report documents a later public MFNet reconstruction using DFormerV2-S. In a matched, single-seed, 20-epoch comparison, RGB-thermal input raised nine-class test mIoU from 49.89% to 51.32%, and nighttime person IoU from 48.90% to 58.20%. The results support a thermal-input gain in this setting; they do not establish superiority over other RGB-T methods or deployment readiness.

1. Context and contribution

My 2025 internship work covered attention-prior and fusion adaptation, data preparation, training, and evaluation for exploratory automotive perception. Company-collected data and historical company results are not included.

All displayed predictions and metrics come from a 2026 public MFNet reconstruction. It reuses DFormerV2's upstream prior formula [1, 2], replacing auxiliary depth with thermal intensity. The complete historical formula and fusion changes were not recovered, so this is not an exact reproduction of the internship implementation.

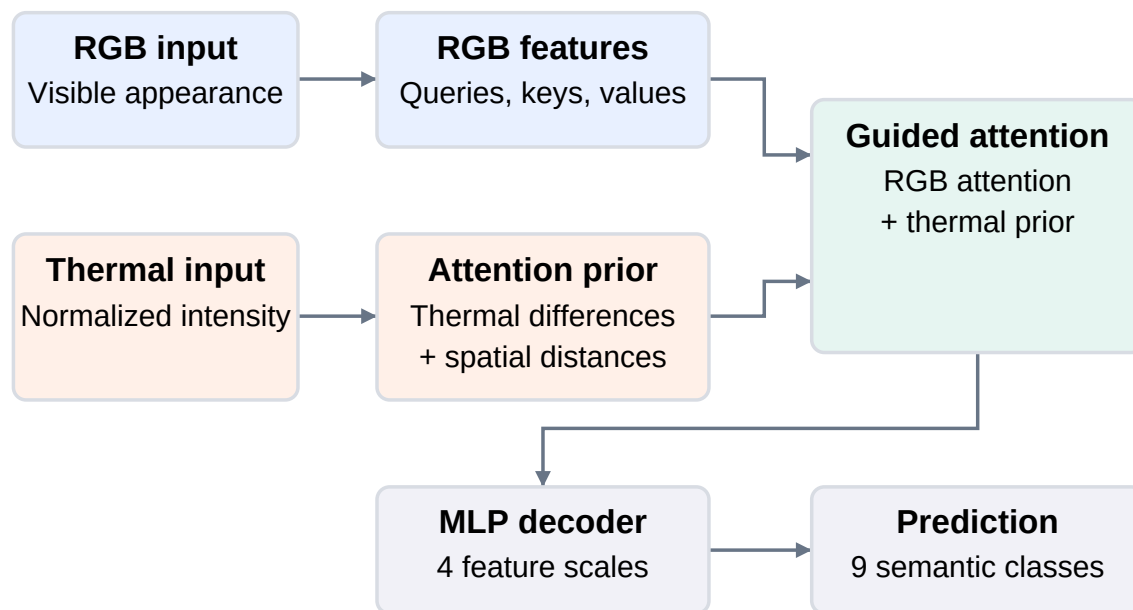


Figure 1. Reconstruction workflow. Thermal input supplies an attention prior; RGB features carry the queries, keys, and values. Four feature scales feed an MLP decoder.

2. Method and experimental setup

Attention guided by thermal intensity

The adapted encoder uses RGB feature tokens for attention. At each scale, pairwise differences in normalized thermal intensity and spatial distance form an additive bias. A compact description of the reused prior is:

$$B_{h,ij} = c_h [w_s (|x_i - x_j| + |y_i - y_j|) + w_t |T_i - T_j|]$$
$$A_h = \text{softmax}(Q_h K_h^T / \text{sqrt}(d_h) + B_h)$$

Here, T is normalized thermal intensity, (x, y) are feature-grid coordinates, c is a head-specific negative decay factor, and w_s and w_t are learned weights. Q and K include the upstream positional transformation. The learned weights are not constrained to be nonnegative; the formula should not be described as always favoring similar temperatures. The first three stages use axial attention, and the last stage uses full two-dimensional attention.

There is no parallel thermal encoder in this reconstruction. Thermal intensity is neither calibrated temperature nor physical depth. RGB-only uses a constant auxiliary map of 0.5 during both training and inference; thermal differences become zero while the spatial prior and parameter shapes remain.

Data handling and training

MFNet images contain RGB in their first three channels and thermal in the fourth channel, which is read as data rather than transparency. Missing pre-flipped filenames are resolved to their original IDs and deduplicated. Original IDs do not overlap across the three splits.

Setting	Matched configuration
Data splits	784 training / 392 validation / 393 test images
Input / output	384 x 512 training input; predictions resized to 480 x 640 before argmax
Normalization	RGB: ImageNet statistics. Thermal: per-image 1st-99th percentile scaling, clipped to [0, 1].
Resizing / augmentation	Bilinear RGB and thermal; nearest-neighbor labels. Shared horizontal flip and RGB brightness augmentation.
Architecture / initialization	DFormerV2-S + 128-channel multiscale MLP decoder. Same NYU-pretrained encoder and decoder initialization seed.
Budget / seed	20 epochs and 3,920 steps per variant; batch size 4; one seed (20261006).
Optimizer / schedule	AdamW: encoder lr 3e-5, decoder lr 8e-4, weight decay 0.01; polynomial lr decay (power 0.9).
Loss / fine-tuning	Class-weighted cross-entropy + 0.3 foreground Dice. Encoder frozen for 2 epochs, then fine-tuned; encoder batch normalization stays in evaluation mode.
Checkpoint selection	Independent best validation foreground mIoU (8 classes): RGB-only epoch 18; RGB-T epoch 15.

Each variant was trained independently with the same data, loss, initialization, and training settings. Training used BF16 autocast and gradient clipping at 1.0. One seed per variant is a controlled pilot, not an estimate of variation across seeds. Bitwise determinism was not guaranteed.

3. Quantitative results

The table below reports saved evaluation results over every test image in each group. Gains are percentage points (pp), computed before rounding. Overall mIoU is derived from the full confusion matrix, not an average of the day and night mIoUs.

Test group / metric	RGB-only (%)	RGB-T (%)	Gain (pp)
All 393: 9-class mIoU	49.89	51.32	+1.43
All 393: Foreground mIoU	43.97	45.56	+1.59
All 393: Person IoU	57.35	63.06	+5.71
Day 205: 9-class mIoU	45.76	46.02	+0.26
Day 205: Foreground mIoU	39.32	39.61	+0.29
Day 205: Person IoU	65.44	67.21	+1.77
Night 188: 9-class mIoU	48.69	51.39	+2.70
Night 188: Foreground mIoU	42.62	45.62	+3.00
Night 188: Person IoU	48.90	58.20	+9.30

IoU = $TP / (TP + FP + FN)$. Nine-class mIoU includes Unlabeled; foreground mIoU averages the other eight classes. Confusion matrices pool all test pixels within each group. Checkpoint selection uses validation foreground mIoU, not test performance.

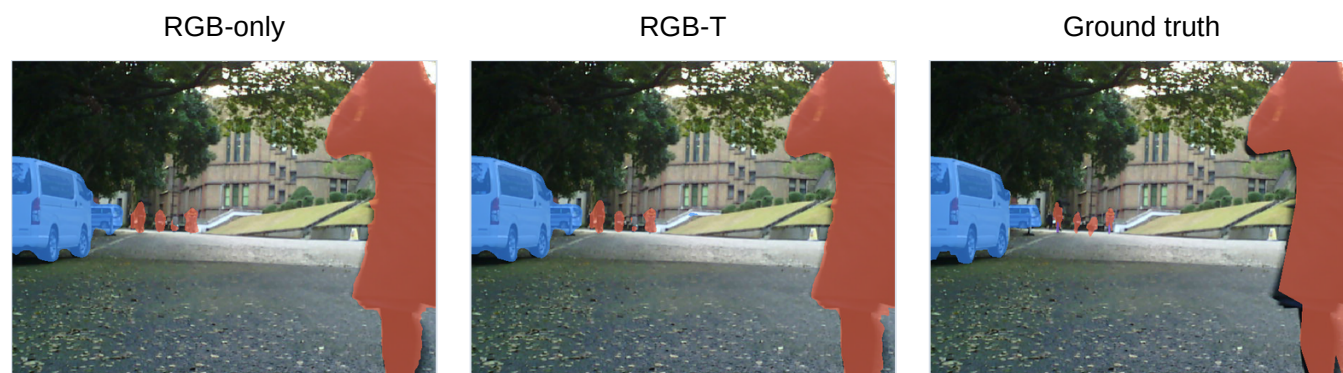
Per-class IoU on all 393 test images

Class	RGB-only (%)	RGB-T (%)	Gain (pp)
Unlabeled	97.27	97.42	+0.16
Car	84.61	85.11	+0.50
Person	57.35	63.06	+5.71
Bicycle	58.16	60.51	+2.35
Curve	35.76	39.08	+3.32
Car stop	26.86	27.57	+0.70
Guardrail	4.13	4.40	+0.27
Color cone	47.33	46.87	-0.46
Bump	37.53	37.87	+0.34

The nighttime person gain (+9.30 pp) is larger than the daytime person gain (+1.77 pp). Gains are not uniform: guardrail IoU remains only 4.40% overall, and Color cone (traffic cone) IoU falls by 0.46 pp. These measurements describe this reconstruction and its RGB-only control; they are not a comparison against other RGB-T architectures.

4. Visual examples and limitations

Daytime - MFNet 01464D



Nighttime - MFNet 01335N



Figure 2. Unretouched predictions and ground-truth overlays from two fixed MFNet scenes. Shared colors: Car (blue), Person (orange), Bicycle (purple), Curve (green). Unlabeled pixels are transparent. Full inputs and enlarged panels are available on the project page.

The scenes were selected for RGB-T car/person outputs before RGB-only results were available; they are not random samples. For 01335N, person IoU rises from 46.89% to 72.96%, but boundaries and legs remain imperfect. In 01464D, car IoU falls from 94.86% to 93.96%. The headline gains come from the full test set, not these examples.

Scope and remaining limitations

This single-dataset, single-seed, 20-epoch pilot does not quantify run-to-run uncertainty or compare alternative RGB-T fusion methods. Small-object and rare-class segmentation remains weak. It does not establish real-time performance, robustness on company data, or deployment readiness. The historical internship implementation is only partially recoverable, so the current prior must be credited to its upstream source.

Evidence used for this report

Prepared from the saved reconstruction code, comparison.json, fixed figure examples, and experiment notes. The report reuses completed runs and saved predictions; no new training or evaluation run was performed to prepare it.

References

- [1] B.-W. Yin, J.-L. Cao, M.-M. Cheng, and Q. Hou. DFormerv2: Geometry Self-Attention for RGBD Semantic Segmentation. CVPR 2025. arxiv.org/abs/2504.04701
- [2] DFormer official implementation. github.com/VCIP-RGBD/DFormer
- [3] MFNet-pytorch: public RGB-thermal dataset and implementation. github.com/haqishen/MFNet-pytorch